



$$U = e^{-\frac{1}{i\hbar}(t-t_0)H}$$

States: $|\psi\rangle_H = U|\psi(t)\rangle$

Heisenberg picture

← S picture

$$= e^{-\frac{1}{i\hbar}(t-t_0)H} \times \left[e^{\frac{1}{i\hbar}(t-t_0)H} |\psi_0\rangle \right]$$

But since $[H, H] = 0$

then, $e^{-\frac{1}{i\hbar}(t-t_0)H} e^{\frac{1}{i\hbar}(t-t_0)H} = \mathbb{1}$

$$e^A e^B = e^{A+B} \Leftrightarrow [A, B] = 0$$

Campbell-Hausdorff formula

$$\Rightarrow |\psi\rangle_H = |\psi_0\rangle \quad \text{NO TIME DEPENDENCY}$$

Operator: $A_H = U A U^\dagger$

← S picture

$$\Rightarrow A_H(t) \text{ picks up a time dependency}$$

Differential eq.?

$$i\hbar \partial_t A_H(t) = i\hbar \left[(\partial_t U) A U^\dagger + U A (\partial_t U^\dagger) \right]$$

but: $\partial_t U = -\frac{1}{i\hbar} H U$

$$(\partial_t U^\dagger) = \left(-\frac{1}{i\hbar} H U \right)^\dagger = \frac{1}{i\hbar} U^\dagger H \quad (H^\dagger = H)$$

Then : $i\hbar \partial_t A_H(t) = \left[-\underbrace{H D A U^\dagger}_{A_H(t)} + \underbrace{U A U^\dagger H}_{A_H(t)} \right]$

$\Rightarrow i\hbar \partial_t A_H(t) = [A_H(t), H]$

d) interaction picture:

let's assume that H can be decomposed into:

$H = H_0 + V$

free hamiltonian

interaction part

Then we choose:

$U = \exp\left[-\frac{1}{i\hbar}(t-t_0)H_0\right]$

states:

$|\psi\rangle_I = U |\psi(t)\rangle$

interaction picture

$= e^{-\frac{1}{i\hbar}(t-t_0)H_0}$

$e^{\frac{1}{i\hbar}(t-t_0)H}$

$= [A, B]$

$|\psi_0\rangle$

$|\psi_0\rangle$

$\Rightarrow |\psi(t)\rangle_I \leftarrow \text{TIME DEPENDENCY}$

Operator : $A_I = U(t) A U^\dagger(t)$

$\Rightarrow A_I(t) \leftarrow \text{TIME DEPENDENCY}$

We can show that (check) $A_I(t)$ is solution of :

$i\hbar \partial_t A_I(t) = [A_I(t), H_0]$ the time evolution of A_I is the one of a free object

Quantum Oscillator (Fly - by ...)

the classical 1D quantum oscillator :

$H = \frac{p^2}{2m} + \frac{1}{2} m \omega^2 x^2$

$[x, p] = i\hbar$

① Introducing dimensionless (dimensionless) variables

$$X = \sqrt{\frac{m\omega}{\hbar}} x$$

$$P = \frac{1}{\sqrt{m\hbar\omega}} p$$

then $H = \frac{1}{2} \hbar \omega (X^2 + P^2)$ with $[X, P] = i$

② We introduce the ladder operators:

$$a^- = \frac{1}{\sqrt{2}} (X + iP) ; \quad a^+ = \frac{1}{\sqrt{2}} (X - iP)$$

with $[a^-, a^+] = 1$

Then, $H = \hbar \omega \left(\frac{1}{2} + a^+ a^- \right)$

We introduce the Hermitian operator $N = a^+ a^-$

$$\Rightarrow [N, a^-] = -a^- \quad \text{and} \quad [N, a^+] = a^+$$

and $[N, H] = 0 \longleftarrow$ there is a common set of eigenstates $\{|n\rangle\}$

i.e. $\begin{cases} N|n\rangle = n|n\rangle \\ H|n\rangle = E_n|n\rangle \end{cases}$ (it is equivalent to use $\{n\}$ or $\{E_n\}$ to describe the states)

★ action of the operator a^- on $a|n\rangle$ state

$$[N, a^-] |n\rangle = N(a^- |n\rangle) - a^-(\underbrace{N|n\rangle}_{n|n\rangle}) \in \mathbb{C}$$

$$N(a^- |n\rangle) = n a^- |n\rangle - a^- |n\rangle = (n-1) (a^- |n\rangle)$$

$\bar{a}|n\rangle$ is an eigenstate of N with eigenvalue $(n-1)$

$$\Rightarrow |n-1\rangle \leftrightarrow \bar{a}|n\rangle$$

i.e. $\bar{a}|n\rangle = \alpha |n-1\rangle$

$\swarrow$ some normalizing factor

Hence $|\alpha|^2 = \langle n-1 | \alpha^* \alpha | n-1 \rangle$

$$= \langle n | a^\dagger \bar{a} | n \rangle \Rightarrow \langle n | \overset{N}{\underbrace{a^\dagger a}} | n \rangle = n \frac{\langle n | n \rangle}{1}$$

$$\Rightarrow |\alpha|^2 = n \quad \leftarrow \quad n \geq 0 \quad \text{real}$$

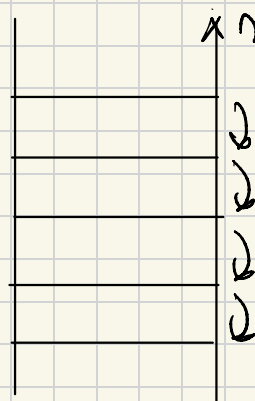
$(\bar{a}|0\rangle = 0)$
 $\swarrow$ vacuum states

$$\Rightarrow \text{let us choose } \alpha \in \mathbb{R}^+$$

Hence $\bar{a}|n\rangle = \sqrt{n} |n-1\rangle$

$\bar{a}$: annihilation operator

(removes 1 unit of n
 1 quantum of energy $\hbar\omega$)



In the same manner:

$$a^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle$$

$\rightarrow$ creation operator

$$\frac{1}{\sqrt{n!}} (a^\dagger)^n |0\rangle = |n\rangle \quad (\text{check this})$$

CHAPTER - 2. Classical Field Theory

I. Continuous systems - classical fields?

a field theory is a theory with a continuum of variables (instead of a set of discrete variables)

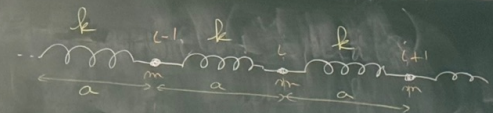
-> From a discrete to a continuous formulation:

Problem set up:

elastic string

↳ infinite chain of point masses attached with identical massless springs

equilibrium



out of equilibrium



The set of $\{q_i, i \in [1, N]\}$ is the set of discrete variables which represents a possible state of the system.

We will assume short range interaction:

$$H = \sum_i \left[\frac{1}{2} m \dot{q}_i^2 + \frac{1}{2} k (q_{i+1} - q_i)^2 \right]$$

It will be useful to factorise 'a':

$$H = a \sum_i \left[\frac{1}{2} \frac{m}{a} \dot{q}_i^2 + \frac{1}{2} k a \left(\frac{q_{i+1} - q_i}{a} \right)^2 \right]$$

Introducing the corresponding lagrangian:

$$L = a \sum_i \left[\frac{1}{2} \frac{m}{a} \dot{q}_i^2 - \frac{1}{2} k a \left(\frac{q_{i+1} - q_i}{a} \right)^2 \right]$$

Equation of motion:

Euler-Lagrange equation:

$$\frac{\partial L}{\partial q_i} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} = 0$$

Show that

$$0 = -\frac{m}{a} \ddot{q}_i - k a \frac{2q_i - q_{i+1} - q_{i-1}}{a^2}$$

Going to the continuum:

This amounts to take $a \rightarrow 0$ in a specific way

$\frac{M}{a} \rightarrow \mu$: mass density of the string

$ka \rightarrow \kappa$: Young's modulus of the string

$i a \rightarrow \chi$: position of the string

$$\varphi_i(t) \rightarrow \varphi(\chi, t)$$

discrete
set of
variables

continuous
set of
variables

Displacement field

↳ mapping of a space - the
point is a physical quantity

$$a \sum_i \rightarrow \int_0^L dx$$

Then: $\varphi_{i+1}(t) - \varphi_i(t) \rightarrow \underbrace{\varphi(x+a, t) - \varphi(x, t)}_{\varphi(x, t) + \frac{\partial \varphi}{\partial x} a + O(a^2)}$

$$\simeq \frac{\partial \varphi}{\partial x} a$$

So, $L = \int dx \left[\frac{1}{2} \mu (\partial_t \varphi)^2 - \frac{1}{2} \kappa \left(\frac{\partial \varphi}{\partial x} \right)^2 \right]$

Lagrangian density $\mathcal{L}$

We have also the action of this system

$$S = \int_{t_1}^{t_2} dt \int_0^L dx \mathcal{L}$$

with $\mathcal{L}(\partial_t \varphi, \partial_x \varphi) = \frac{1}{2} \mu (\partial_t \varphi)^2 - \frac{1}{2} \kappa (\partial_x \varphi)^2$

By taking the continuum limit of the discrete eq. of motion we get (exercise):

$$-\mu \left(\frac{\partial \varphi}{\partial t} \right)^2 + \kappa \left(\frac{\partial \varphi}{\partial x} \right)^2 = 0$$

Comments :

1- in general, $\mathcal{L}(\partial_t \varphi, \partial_x \varphi; t)$

2 - discrete system with short range interaction $\xrightarrow[\text{limit}]{\text{continuum}}$ $\mathcal{L}$ $\left(\begin{array}{l} \text{first order} \\ \text{derivatives of} \\ \text{the field with} \\ \text{respect to} \\ \text{time and space} \end{array} \right)$

3- the corresponding action is LOCAL because S is an integral over time and space of a Lagrangian density which depends on the values of the field and its derivatives on one point. Moreover, the Lagrangian density is a polynomial.

Remarks:

★ The Lagrangian formalism will be mainly used because it treats time and space on the same footing (relativity)

★ The hamiltonian relies on time explicitly (space is disentangled)

• $\varphi(x, t)$:

↖ ↗ continuous variables

↘ dynamical variable (continuous)